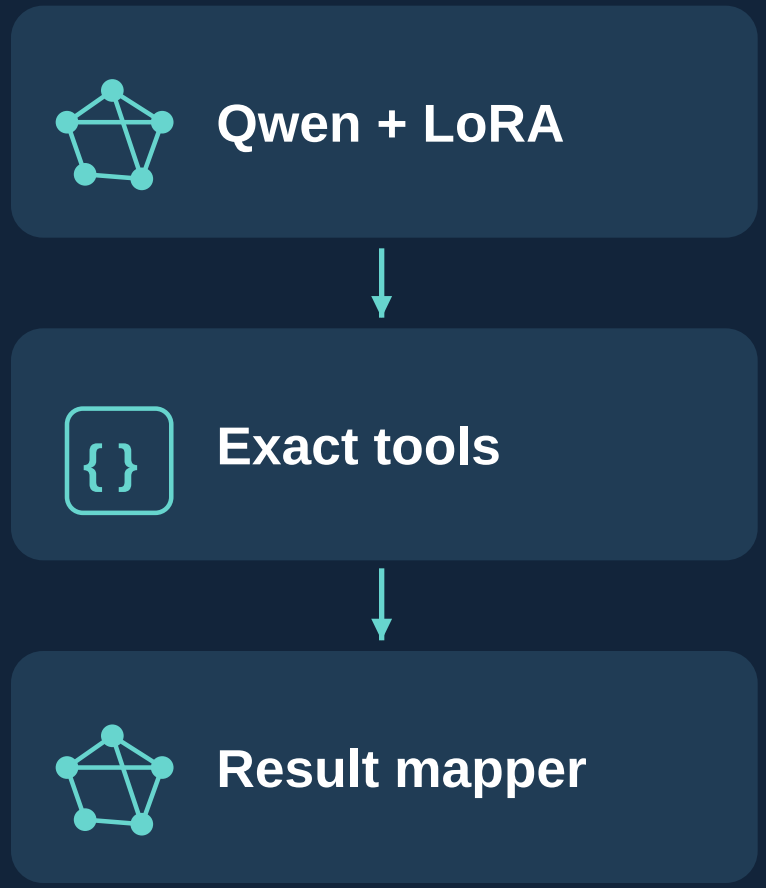
PROJECT REPORT

LLM Character Operations

Understand the request.
Compute exactly.
Feed the result back.

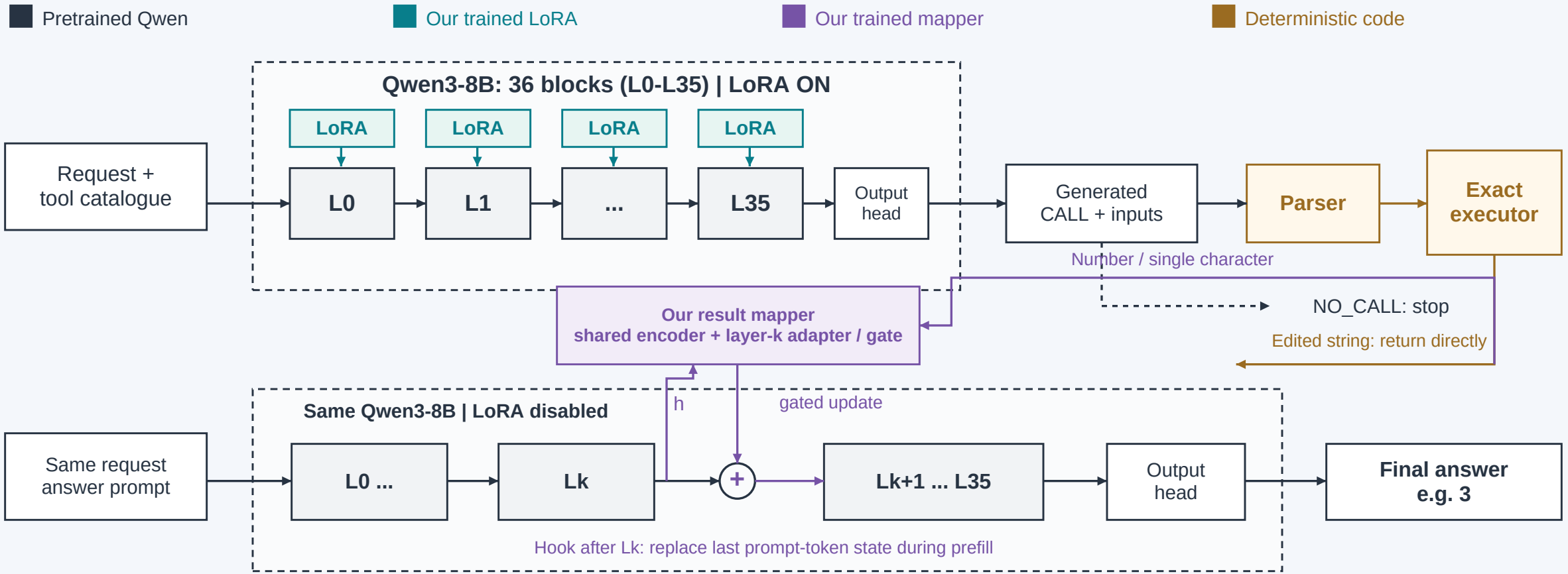


Qwen3-8B backbone | approximately 8 billion parameters

Report to Professor | 15 October 2026

Architecture of the inference pipeline

Separate pretrained backbone, trained modules and deterministic computation.

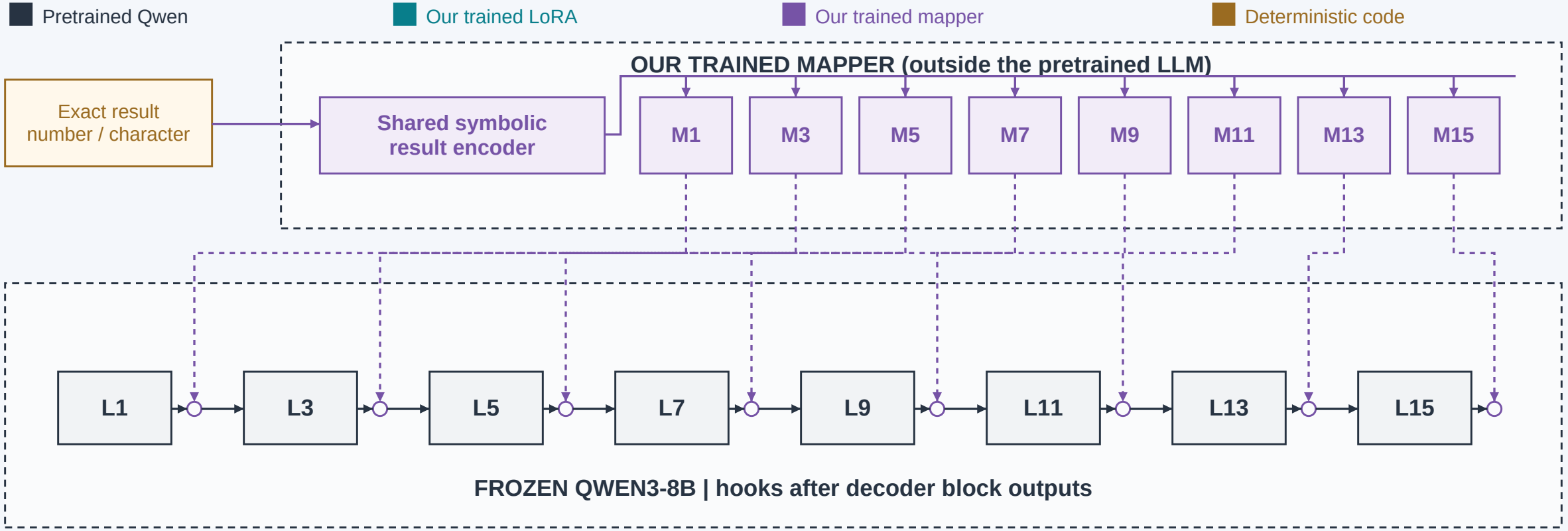


LoRA = small adapters inside Qwen projections, not a separate stack of layers.

Two generation passes. No ground-truth correction of the predicted call.

Layer-wise result injection

Shared result encoding, separate adapters and one selected decoder-output hook.



Even-numbered blocks and later blocks are omitted.

Train across 8 candidates / select on development data / use one hook per answer

Dashed paths are alternatives. Initial three-operation run chose zero-based L5.

Our dataset: CWUM-inspired

Custom English data inspired by Zhang & He's CWUM paper (EMNLP 2024).

50k original / 150k expanded

12 balanced operations

63% tool tasks

37% no-call controls

Count / measure

Characters, text length, words, word length

Find / retrieve

Character position; character in text or word

Edit / reverse

Insert text or words; reverse text or one word

30% real words

50% modified words

20% random strings

CWUM-style prompts + our own templates; exact answers generated by code.

Separate train / dev / test / new-wording test. Expanded data adds longer sentences.

CWUM paper

Fresh generated examples; no copied benchmark test items. Tools are allowed in our protocol.

How our pipeline differs

The key change is how a computed number or character returns to the model.

Ordinary tool feedback

Our scalar-result route

Proposed novelty / contribution

A learned, layer-specific gated mapper reintegrates exact numeric and single-character tool results into frozen Qwen3-8B, paired with a LoRA-trained selector.

Current scope: edited strings return directly. Hidden-state feedback broadly has prior work.

Matched text-feedback experiments must establish the mapper's benefit.

Small-model results and our progress

Our Qwen3-8B: ~8B parameters, about 9x smaller than 70-72B models. Different benchmarks below.

Study / system	Task / setting	Reported result
CWUM (2024) [1]	English task average	Llama-3-8B: 25.13%; 7B SFT model: 93.71%
CharBench (2025) [2]	Counting + indexing	Mistral-7B: 34.72% overall
Divide & Conquer (2025) [3]	Deletion / insertion / substitution	Llama-3-8B: 72.2% / 63.8% / 73.2%
Our initial Qwen3-8B + mapper	3 operations; final task answer	IID 99.62% (2334/2343); new wording 98.11% (883/900)
Our newer Qwen3-8B selector	12 operations + controls; call exact	Test 99.88%; held-out 94.26% (4713/5000)
Current 12-operation pipeline	LoRA + multilayer-candidate mapper	Completed full-pipeline score not supplied

Initial run: count character, string length, find character; reused V6 mapper at L5. Not the current 12-task protocol.

Initial overall success incl. controls: 99.78% (3991/4000) IID; 96.89% (1744/1800) new wording.

Different tasks and training: not a direct ranking. Initial: copied JSONs; newer selector: reported server logs.

[1] CWUM, Tables 2 and 4 [2] CharBench, Table 2 [3] Divide & Conquer, Table 1